THE SCIENCE

Formulas, Notation & Logic

Every number the system reports, derived. We measure the way the 2025 conversation-intelligence and LLM-evaluation literature does — faithfulness and speaker separation — not a brittle exact-match "accuracy".

1 · Diarization — channels, not guessing

Let `L` = agent channel, `R` = customer channel. We transcribe each separately into turn sets `T_L` and `T_R`, then rebuild the real conversation:

```
T = sort( T_L ∪ T_R )   ordered by turn.start
```

Separation quality

To prove the two channels really are separate speakers, take the Pearson correlation `ρ` between the two channels' energy envelopes `e_L, e_R` (100 ms windows):

$$\rho = \frac{\Sigma\,(e\_L_i - \bar{e}\_L)(e\_R_i - \bar{e}\_R)}{\sqrt{[\ \Sigma(e\_L_i - \bar{e}\_L)^2 \cdot \Sigma(e\_R_i - \bar{e}\_R)^2\ ]}}$$

```
separation_quality  =  1 − max(0, ρ)
```

Low (even negative) `ρ` ⇒ the speakers don't overlap ⇒ attribution is correct by construction. Measured on this dataset: `ρ ≈ −0.13` → `separation_quality = 1.00 (100%)`.

ॐ विघ्नहर्ता

→ Formulation & Notations :-

Diarization - my channels, not guessing

let
'L' → agent
'R' - Customer

we transcribe each separately, giving term sets like 'T_L' and 'T_R'

So transcript the final one is like

$$T = \text{Sort}\,(T\_L \cup T\_R) \text{ order by turn. start}$$

→ separation quality :-

let me prove the channels really are separate speaker by the person Correlation 'ρ' between the two channels' energy envelops "e_L", "e_R" (100ms window):-

Hand-worked: diarization by channels, and the transcript reconstruction.

so formulation:-

$$\rho = \frac{\sum (c\_L_i - \bar{c}\_L)(c\_R_i - \bar{c}\_R)}{\sqrt{\left[\sum (c\_L_i - \bar{c}\_L)^2 \cdot \sum (c\_R_i - \bar{c}\_R)^2\right]}}$$

$$\text{separation\_quality} = 1 - \max(0, \rho)$$

Low (even negative) 'ρ' ⇒ the two speaker don't overlap ⇒ attribution is correct ** by construction"

Measured on this dataset that given

$$\rho \approx -0.13$$

separation_quality = 1.00 (100%) nearly.

→ Evidence verification -

normalise(s) = lowercase(s) with only [a-z0-9]

$$\text{verified}(e\_j) = 1 \text{ if normalise(quote)} \subseteq \text{normalize}(z.\text{text}) \text{ or } |Q \cap B| / |Q| \geq 0.60$$

0 otherwise

where Q = words(quote), B = words(z.text)

Hand-worked: Pearson ρ, separation quality, and the evidence-verification rule.

2 · Evidence verification & faithfulness

Each judgment `j` carries evidence `e_j = (timestamp, quote, speaker)`. Verification finds the transcript turn `τ` at that timestamp+speaker, normalises both, and accepts if the quote appears in the turn *or* shares ≥ 60% of its significant words:

```
normalise(s) = lowercase(s) keeping only [a-z0-9 ]

verified(e_j) = 1  if  normalise(quote) ⊆ normalise(τ.text)
                   or  |Q ∩ B| / |Q|  ≥  0.60
                0  otherwise      (Q = words(quote), B = words(τ.text))
```

Per-call evidence score (n = judgments checked):

```
evidence_score = (1/n) · Σⱼ verified(e_j)
```

System-wide faithfulness (over all calls C):

```
                Σ_c Σ_j verified(e_{c,j})
faithfulness = ─────────────────────────  = 0.999  (99.9%)
                  Σ_c (# judgments in c)
```

Per call evidance score.

$$\text{evidance-score} = \frac{1}{n} \times \sum_j \text{verifie}(e_j)$$

System wide faithfulness (over all calls c);

$$\text{Faith} = \frac{\sum_c \sum_j \text{verified}(e_{\{c,j\}})}{\sum_c (\#\ \text{judgmens in } c)}$$

$$= 0.999 \quad (99.9\%)$$

Attention Score (Need)

$$A \in [0, 100]$$

Hand-worked: per-call evidence score, system-wide faithfulness, and the attention score.

3 · Needs-attention score A ∈ [0, 100]

Additive signals, then clamped. `neg` = # negative cues from the customer.

```
A = 10                                      (base)
  + min(35, 12·neg)                         (customer negativity)
  + 30 escalated / 28 unresolved / 15 unclear
  + 20 if mood shifted to negative  (−5 if recovered to positive)
  + 22 if HIGH_RISK topic / 10 if MED_RISK
  + 12 if agent repeated a question
  + 10 if dur ≥ 90s  (5 if ≥ 70s)
A = clamp(A, 0, 100)

HIGH_RISK = {fraud_dispute, card_lost_stolen, complaint, account_access}
MED_RISK  = {transaction_issue, fees_charges, loan_mortgage, payment_transfer}
```

4 · QA / compliance score Q ∈ [0, 100]

Each check `k` has a pass flag `p_k ∈ {0,1}` and a weight `w_k`:

CHECK	WEIGHT
Greeting (opened professionally)	1
Identity verified before a sensitive action	3 (0 if not required)
Action confirmed	2
Empathy shown	1
No repeated question	1
Proper close	1

```
      Σ_k  p_k · w_k
Q =  ─────────────── × 100        (over checks with w_k > 0)
        Σ_k  w_k

resolution_risk =
   (sensitive_action ∧ ¬identity_verified)                    # compliance risk
 ∨ (sounded_resolved ∧ customer_asked ∧ ¬action_confirmed ∧ status ≠ resolved)
```

QA | Compliance score $Q \in [0, 100]$

$$Q = \frac{\sum_k p_k \cdot w_k}{\sum_k w_k} \cdot 100$$

over checks with $w_k > 0$

Number of quote (full run 1441 calls)

Parameter	Score
calls	1441
customers	100
agents	10
faithfulness	99.9%
diarization	100%
coverage	100%
resolution risk	195

Hand-worked: the QA score, and the measured results on all 1,441 calls.

Measured results (full 1,441-call run)

PARAMETER	VALUE
Calls	1,441
Customers	100
Agents	10
Faithfulness	99.9%
Diarization separation	100%
Coverage	100%
Resolution-risk calls	195